### 4.2.1 Transversal

A line which intersects two or more lines at distinct points is called a transversal of those lines.

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Case (i) When a transversal intersect two lines, we get eight angles.

In the figure the line $l$ is the transversal for the lines $m$ and $n$

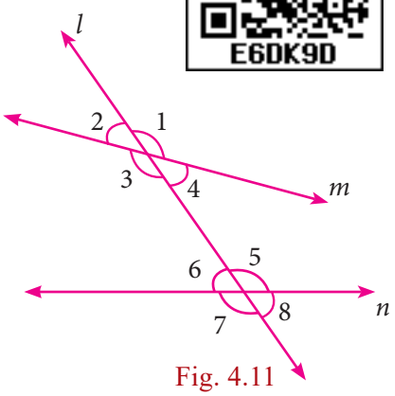



Fig. 4.11

(i) Corresponding Angles: $\angle 1$ and $\angle 5$, $\angle 2$ and $\angle 6$, $\angle 3$ and $\angle 7$, $\angle 4$ and $\angle 8$

(ii) Alternate Interior Angles: $\angle 4$ and $\angle 6$, $\angle 3$ and $\angle 5$

(iii) Alternate Exterior Angles: $\angle 1$ and $\angle 7$, $\angle 2$ and $\angle 8$

(iv) $\angle 4$ and $\angle 5$,$\angle 3$ and $\angle 6$ are interior angles on the same side of the transversal.

(v) $\angle 1$ and $\angle 8$, $\angle 2$ and $\angle 7$ are exterior angles on the same side of the transversal.

Case (ii) If a transversal intersects two parallel lines. The transversal forms different pairs of angles.

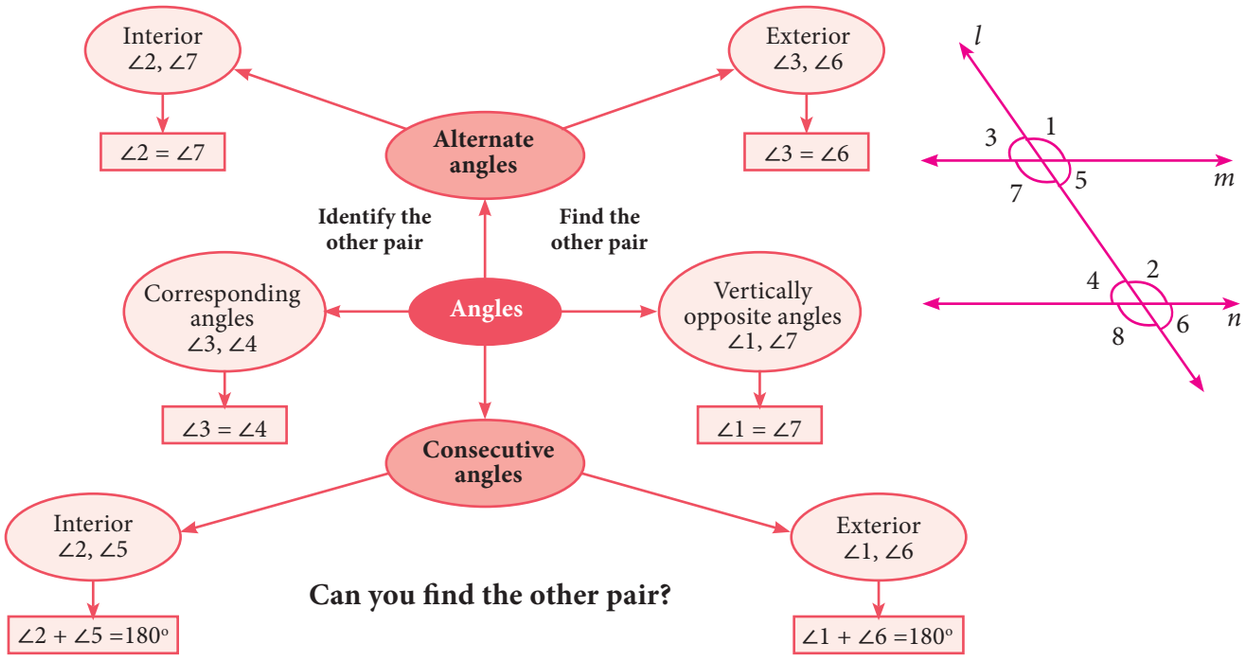



Fig. 4.12



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